

Advanced Data Analysis



These are the same subjects measured before (pre_score) and after (post_score) an intervention. Did their scores change significantly?

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Yes. Because the same subjects were measured before and after the intervention, the appropriate analysis is a **paired-samples t-test**.

Results

- Number of subjects: **25**
- Mean change (post – pre): **4.32 points**
- Test statistic: **$t(24) = 2.83$**
- p-value: **0.009**

Since the p-value is below 0.05, the change is statistically significant.

Interpretation

The intervention was associated with a **significant increase in scores** from pre-test to post-test.

The estimated average improvement was about **4.3 points** (95% CI: 1.17 to 7.48).

Effect size:

- Cohen's $d \approx 0.57$, which is typically interpreted as a **moderate effect**.



